

GAME 10017 Game Systems Design

Assignment 1: Tetris Puzzle - By Jake Kielty

1. What interesting properties does your polyomino piece have within the systems of Tetris? Why?

My piece combines a T and an I, which allows the player to use it to fill one space gaps like they would with a T, but can also fill two space gaps with the long side of the T. It is essentially a T with the added consideration of it being one space longer. Additionally, because of its 4-tile length, it can be used to fill out longer spaces, provided that there is space to work around the extra tile sticking out.

This shape is generally somewhat awkward to fit in, though, and so it should provide a little bit of an additional challenge than a more simplistic shape like a 2x3 or a 1x5 (or larger) rectangle would. Shapes like the O and the I are generally pretty easy to use, and quite “safe” in that one can make 1 or 2 tile gaps that will ensure that you can clear some lines when you get the correct blocks. As long as there is an opening, you can pretty consistently use these two polyominoes to clear up lines in situations where others would leave gaps or leftover bits. Knowing this, I wanted to avoid making a shape like this that would be too easy to fit in.

On the other hand, it is not such an irregular shape that it can only fit into very obvious spots. If the shape were something much larger or with a very irregular shape, then you’d have to leave much larger and more obvious spaces in the board to make it work, which would make it easier for the player to figure out where it goes. If the piece is some 8-pointed star-like shape, then you would know that it’s only really going to fit into a spot where all of those points can slot in. Similarly, if it’s a large 4x4 square, then there will need to at least be a 4 tile wide channel above where it’s supposed to go, which would make it incredibly obvious where the player should place it.

As opposed to very simple or very complex/large pieces, my piece sits somewhere between those options, and provides some options for clearing like the I piece does, while also requiring some more consideration than it does. Additionally, because the piece is not *too* complex, it doesn’t immediately give away exactly where it must be placed, and gives some room for the player to have to consider where they will place it.

2. What interesting relationships does your piece sequence have with the board state within the envisioned play session? Why?

Generally speaking, my piece is a fairly simple tweak to an existing piece, and so it is intended to be familiar, but different enough that the player cannot just simply apply it in the same way they'd apply an I or a T. Within the piece sequence, as it is the first piece that the player will see, it will immediately show the player that all of the tiles placed will be factored into the later row clears, as the bottom-most square of the piece fills a gap that will allow the player to clear line 15 with the third piece, which is a T.

Later on, as the sixth piece in the sequence, my piece is used in a more conventional manner, where it is used to clear rows 12 and 13 together. As for how this all fits into the rest of the sequence, it is about in line with how most of the rest are used: an opening is provided but there are also potentially one or two other spots where the player could choose to slot them in (that may not clear lines). Now, a player that *knows* they must clear at least one line per piece would probably be able to see where these slot-in relatively clearly, but unfortunately, I couldn't quite get things to a spot where there could be two possible line clears per turn with only one of them being correct for the sequence.

Overall, my piece sequence is meant to have some variety and is intended to avoid the cliché "leave a space on one of the sides for an I piece" strategy that players often use in Tetris. There is one I piece, and it *is* used in a very similar manner, but that is meant to be a part of the faster paced and easier latter half of the board.

3. How do you expect your board state to provide an interesting challenge to players?

My board is intended to provide some level of challenge by having a starting board state that does not immediately give away each move. I think it will be interesting because it will start off with a clear end point (the final few moves are very simple) but will provide a much less clear starting point, where the player will soon realize where their various pieces are intended to slot in.

I think it will be interesting because once the player can get to the last few steps, they'll be able to have a sort of "aha!" moment where they'll be able to quickly slot in the last few and clear the board in a satisfying manner. The early pieces are meant to make the player consider their placement a bit more, and should provide at least *some* challenge, where the later pieces are meant to be more simple to place.

I think this should be a relatively easy, but not brainless board to clear. I am far from experienced in this type of Tetris, and don't really know how to make something that adheres to the rules laid out while also being very difficult, so I shot for something with a low-moderate difficulty that someone like myself could clear without it being annoying.

4. Challenges faced during implementation and how they were addressed.

(Just added this because what I went with diverges from what I originally wrote here)

Challenges-wise, I found the implementation of my piece and the intended gameplay mechanic to be fairly smooth, although I could not figure out how to get the more complex features working (hence why I went with a time trial). I did want to do the same type of “Special Board State” challenge as was in this document, but I couldn’t figure out how to get the pieces to spawn in the correct locations *and* clear correctly as though they had already been dropped there.

Instead, I decided to go with a time trial mechanic, where the player has 120 seconds to get as high of a score as possible. For this one, the main challenge was actually getting the UI text to update. For whatever reason, methods in UIController didn’t actually get called each frame when I put them in Board’s update function, so while I originally had it set up like the scoreText, it still wouldn’t get called each frame when I called `uiController.UIUpdateTimer()`. Instead, I decided to put the entire bit of timer code in TetrisManager, since I knew it was already being called each frame, and it worked!

It’s not exactly as elegant as I’d like, but the alternative was having it run an event for it every frame, and I’m not sure if that will cause issues or not, so I’d rather keep it simple. The scoreText only needs to be updated when you clear a line, and so its event doesn’t need to be run constantly, but updating timer text would need to be run all the time.

It also generally just took a bit of fiddling about to see what I needed or didn’t need, and I managed to eliminate an unnecessary timeRemaining variable by just subtracting `Time.DeltaTime` from `currentTime` directly.

I found using events and having a modular structure like this to be very convenient, as it meant that I didn’t have to do much beyond adding `SetGameOver(true)` to start the whole game over process. In another context, I’d usually just have it load a separate scene with a game over screen and whatnot (that way it would reset the entire game again by loading the main scene again when they restart), but I found this to be a hell of a lot more convenient.

Besides that stuff, I added some music (which I know wasn’t important here), and that took a little bit of fiddling with as well to get working. Because TetrisManager was already accessible by most classes, I just put the start and stop functions in there, and then had them called at the appropriate locations (`board.ResetBoard()` for when the game starts/restarts, and `UIController.UpdateGameOver()` for stopping it when the game ends).

Link to Video:

<https://youtu.be/DAPrUh0sVfg?si=gmlGnepYj4CV97TQ>

